

Making the Zini–Wang comparison precise

A note on §1.2 of *Conformal field theory from lattice fermions*

31 July 2026

Abstract

Section 1.2 of the published article, and of the companion manuscript `free_fermion_cft_v5.tex` as it stood when this note was written, compares the operator-algebraic renormalization (OAR) construction of the scaling limit with the *low-energy scaling limit* of Zini and Wang, and asserts that the Zini–Wang structure “is not immediately available” for the massless chiral lattice fermions treated there, “where the finitely renormalized states $\omega_M^{(N)}$ are not pure (due to the entanglement of the chiral halves)”. That assertion is stated without proof and, as it stands, identifies the wrong obstruction.¹ We make the comparison precise. The renormalization flow acts on one-particle symbols by *compression* along an isometry, and we give the exact identity governing when a compression of a projection is again a projection (Lemma 2.1). We then observe that what the Zini–Wang connecting unitaries actually require is not purity but constancy of the GNS multiplicity, which for a quasi-free state is $2^{m(S)}$ with $m(S)$ the number of eigenvalues of the symbol strictly between 0 and 1 (Proposition 3.1). Computing m in the model of the manuscript gives closed forms in every case, and yields a sharper statement: for the chiral algebra the multiplicity is *constant in M* , so the connecting unitaries do exist at every finite scale, and the obstruction is a *discontinuity at $M = \infty$* , where the multiplicity collapses by a factor $2^{|\Gamma_N, -|}$ (Theorem 4.1, Corollary 5.1). Quantitatively, the chiral restriction of the finite-scale vacuum has extensive entropy with density exactly $2 \ln 2 - 1$ per site (Corollary 5.2). All closed forms are confirmed numerically in `numerics/os_check19.py`.

1 Introduction and conventions

References to the main text. Throughout, citations of the form §1.2, (60), Definition 3.16, Lemma 4.28 refer to the companion manuscript

`free_fermion_cft_v5.tex` (the revised version of [3]),

with the numbering of that file as compiled on 22 September 2026; the numbers are pulled from its `.aux` into `mainrefs.tex` by `sync_docs.py`, so they stay correct when the manuscript is edited. Where a statement of the manuscript is quoted verbatim it is set in quotation marks. Numerical checks live in `numerics/os_check19.py` of the same directory and are labelled [31]–[34] there.

What §1.2 asserts. The manuscript recalls the Zini–Wang *low-energy scaling limit* [5]: a nested family of energy-bounded subspaces $\mathcal{H}_N^E \subset \mathcal{H}_N^{E'}$, *connecting unitaries* $\phi_N^E : \mathcal{H}_N^E \rightarrow \mathcal{H}_{N+1}^E$ defined for N large enough that $\dim \mathcal{H}_N^E = d_E$ stabilises, and the extension property $\phi_N^E = \phi_N^{E'}|_{\mathcal{H}_N^E}$. It then explains that OAR reproduces the Zini–Wang compatibility condition through the GNS isometries induced by the refining morphisms, and continues:

¹Every quotation of §1.2 in this note is from the published article, Commun. Math. Phys. **398** (2023) 219–289, whose wording the manuscript carried unchanged until 22 September 2026. On that date §1.2 of `free_fermion_cft_v5.tex` was rewritten in the light of this note, and now states the multiplicity criterion of §3 instead; the passages quoted below are therefore no longer to be found there.

“... if the GNS representations $\{\mathcal{H}_M^{(N)}, \pi_M^{(N)}, \Omega_M^{(N)}\}$ of the finitely renormalized states $\omega_M^{(N)}$ are unitarily equivalent, e.g. assuming a Stone–von Neumann-type result at finite scales, that is, if there are (connecting) unitaries $\phi_M^N : \mathcal{H}_M^{(N)} \rightarrow \mathcal{H}_{M+1}^{(N)}$, we can recover the structure of a low-energy scaling limit ... such that N plays the rôle of the energy scale E .”

and, a page later,

“It is worth pointing out that such a structure is not immediately available for the operator-algebraic renormalization of (free) massless chiral lattice fermions in $1 + 1$ -dimensions considered here, where the finitely renormalized states $\omega_M^{(N)}$ are not pure (due to the entanglement of the chiral halves) in contrast with their scaling limit (given by the Hardy projection).”

Two things are left open. First, the hypothesis “assuming a Stone–von Neumann-type result at finite scales” is not verified, nor is it said what it should mean. Second, the non-purity of $\omega_M^{(N)}$ is asserted, not proved, and the inference from non-purity to the unavailability of the Zini–Wang structure is not justified. Sections 2–5 settle all three points. Section 8 records what this note does *not* settle.

Setting. We work with the OAR data of §3 of the manuscript. At each scale N we have a finite-dimensional one-particle space $\mathfrak{h}_N$ ($\dim \mathfrak{h}_N = d_N < \infty$), the CAR algebra $\mathfrak{A}_N = \mathfrak{A}_{\text{CAR}}(\mathfrak{h}_N)$, and isometries $R_{N+1}^N : \mathfrak{h}_N \rightarrow \mathfrak{h}_{N+1}$ forming an inductive system (eq. (87) of the manuscript), with associated Bogoliubov morphisms $\alpha_{N+1}^N : \mathfrak{A}_N \rightarrow \mathfrak{A}_{N+1}$. The renormalization group flow on states is (89),

$$\omega_M^{(N)} = \omega^{(N+M)} \circ \alpha_{N+M}^N, \quad N, M \in \mathbb{N}_0, \quad (1)$$

a state on the *fixed* algebra $\mathfrak{A}_N$, and the scaling limit is $\omega_\infty^{(N)} = \lim_M \omega_M^{(N)}$ (90). Recall that a quasi-free state ω_S on $\mathfrak{A}_{\text{CAR}}(\mathfrak{h})$ is determined by its symbol $S = S^* \in B(\mathfrak{h})$ with $0 \leq S \leq \mathbb{1}$ via $\omega_S(a^\dagger(\xi)a(\eta)) = \langle \eta, S\xi \rangle$, and that ω_S is pure if and only if S is a projection [1, 2]. Since all $\omega^{(N)}$ in the manuscript are quasi-free and the α ’s are Bogoliubov, (1) reads at the one-particle level

$$S_M^{(N)} = (R_{N+M}^N)^* S^{(N+M)} R_{N+M}^N, \quad (2)$$

that is, *the renormalization of a quasi-free state is the compression of its symbol along an isometry*. Everything below follows from this one observation.

Remark 1.1 (Two families of projections). The manuscript uses $\pm$ in two unrelated ways, and both occur below. We write $p_\pm = \frac{1}{2}(\mathbb{1}_2 \pm \sigma_y)$ for the *chirality* projections on the spinor index $\mathbb{C}^2$ of (61), which are constant in the momentum k ; and $P^\pm$ for the *Hardy* projections onto positive respectively negative momenta, which act on the momentum variable and are trivial on the spinor index. The scaling-limit symbol (129) happens to couple the two, $P_0(k) = p_{\text{sign}(k)}$, which is precisely why the chiral scaling limit is pure.

2 Compressions of quasi-free symbols

Lemma 2.1 (Compression identity). *Let $\mathfrak{h}', \mathfrak{h}$ be Hilbert spaces, $R : \mathfrak{h}' \rightarrow \mathfrak{h}$ an isometry, and $S = S^* \in B(\mathfrak{h})$ with $0 \leq S \leq \mathbb{1}$. Put $S' := R^*SR$ and $Q := \mathbb{1} - RR^*$, the projection onto $(\text{ran } R)^\perp$. Then $0 \leq S' \leq \mathbb{1}$ and*

$$S' - S'^2 = R^*(S - S^2)R + X^*X, \quad X := QSR. \quad (3)$$

*In particular, if S is a projection then $S' - S'^2 = X^*X \geq 0$, and the following are equivalent:*

(i) S' is a projection, i.e. $\omega_{S'}$ is pure;

(ii) $X = QSR = 0$;

(iii) $\text{ran } R$ is invariant under S .

Moreover $\|S' - S'^2\| = \|X\|^2$, and if R is the inclusion of a subspace $\mathfrak{h}' \subset \mathfrak{h}$ with orthogonal projection p , then $X = (\mathbb{1} - p)Sp$ and $\|X\| = \|[S, p]\|$. Finally, if S is strictly between 0 and 1, i.e. $0 < S < \mathbb{1}$, then so is S' , and hence

$$m(S') = \dim \mathfrak{h}' \quad \text{whenever } 0 < S < \mathbb{1}, \quad (4)$$

with m as in Proposition 3.1.

Proof. $0 \leq S' \leq \mathbb{1}$ is immediate from $R^*R = \mathbb{1}$. Using $RR^* = \mathbb{1} - Q$,

$$S'^2 = R^*SRR^*SR = R^*S(\mathbb{1} - Q)SR = R^*S^2R - R^*SQSR,$$

and $Q = Q^* = Q^2$ gives $R^*SQSR = (QSR)^*(QSR) = X^*X$; subtracting from $S' = R^*SR$ yields (3). If $S^2 = S$ the first term drops. Then S' is a projection iff $X^*X = 0$ iff $X = 0$, and $X = QSR = 0$ says exactly that $S(\text{ran } R) \subset \text{ran } R$. The norm identity is $\|X^*X\| = \|X\|^2$. For the last statement take $\mathfrak{h}' \subset \mathfrak{h}$ and R the inclusion, so $RR^* = p$ and $X = (\mathbb{1} - p)Sp$; since $[S, p] = (\mathbb{1} - p)Sp - pS(\mathbb{1} - p)$ is the difference of the two off-diagonal blocks of S in the decomposition $\mathfrak{h} = p\mathfrak{h} \oplus (\mathbb{1} - p)\mathfrak{h}$, and these are adjoint to one another, $\|[S, p]\| = \|(\mathbb{1} - p)Sp\|$. For the last claim, if $0 < S < \mathbb{1}$ then for $0 \neq \xi \in \mathfrak{h}'$ we have $\langle \xi, S'\xi \rangle = \langle R\xi, SR\xi \rangle \in (0, \|R\xi\|^2) = (0, \|\xi\|^2)$, because $R\xi \neq 0$; so no eigenvalue of S' equals 0 or 1, which is (4). $\square$

Lemma 2.1 covers both mechanisms that occur in the manuscript with a single statement, because both the renormalization maps R_{N+M}^N and the passage from the full to the chiral algebra are compressions along isometries: the latter is the inclusion of $\mathfrak{h}_{N,\pm}^{(-)} = \ell^2(\Lambda_N)_{\pm} \otimes p_- \mathbb{C}^2$ into $\mathfrak{h}_{N,\pm} = \ell^2(\Lambda_N)_{\pm} \otimes \mathbb{C}^2$ of (61), with $p_{\pm} = \frac{1}{2}(\mathbb{1}_2 \pm \sigma_y)$.

3 What the connecting unitaries actually require

The manuscript's hypothesis is the existence of unitaries $\phi_M^N : \mathcal{H}_M^{(N)} \rightarrow \mathcal{H}_{M+1}^{(N)}$ between the GNS spaces of $\omega_M^{(N)}$ and $\omega_{M+1}^{(N)}$, intertwining the two representations of $\mathfrak{A}_N$. Because $\mathfrak{A}_N$ is finite-dimensional this is decided by a single integer.

Proposition 3.1 (GNS multiplicity). *Let $\mathfrak{A} \cong M_d(\mathbb{C})$ and let ω be a state on $\mathfrak{A}$ with density matrix ρ . Then the GNS representation of ω is unitarily equivalent to $\text{rank}(\rho)$ copies of the unique irreducible representation of $\mathfrak{A}$, so*

$$\dim \mathcal{H}_{\omega} = d \cdot \text{rank}(\rho), \quad (5)$$

and the GNS representations of two states ω, ω' are unitarily equivalent if and only if $\text{rank}(\rho) = \text{rank}(\rho')$. If moreover $\mathfrak{A} = \mathfrak{A}_{\text{CAR}}(\mathfrak{h})$ with $\dim \mathfrak{h} = d_{\mathfrak{h}} < \infty$ and $\omega = \omega_S$ is quasi-free with symbol S , then

$$\text{rank}(\rho_S) = 2^{m(S)}, \quad m(S) := \#\{\text{eigenvalues of } S \text{ in } (0, 1)\} \quad (6)$$

counted with multiplicity, so that $\dim \mathcal{H}_{\omega_S} = 2^{d_{\mathfrak{h}}} \cdot 2^{m(S)}$.

Proof. Every representation of $M_d(\mathbb{C})$ is a multiple of the unique irreducible one on $\mathbb{C}^d$, and two such are unitarily equivalent iff the multiplicities agree; the multiplicity of the GNS representation of ω is $\text{rank}(\rho)$, since $\mathcal{H}_\omega = \mathfrak{A}\rho^{1/2}$ with the Hilbert–Schmidt inner product, and $\mathfrak{A}\rho^{1/2} \cong \mathbb{C}^d \otimes \text{ran}(\rho)$ as a left $\mathfrak{A}$ -module. For the quasi-free case, diagonalise S with eigenvalues $\lambda_1, \dots, \lambda_{d_h}$; in the corresponding mode basis the density matrix factorises, $\rho_S = \bigotimes_i \begin{pmatrix} \lambda_i & 0 \\ 0 & 1-\lambda_i \end{pmatrix}$, whose rank is $\prod_i \text{rank} \begin{pmatrix} \lambda_i & 0 \\ 0 & 1-\lambda_i \end{pmatrix}$; each factor has rank 2 if $\lambda_i \in (0, 1)$ and rank 1 if $\lambda_i \in \{0, 1\}$. Finally $\mathfrak{A}_{\text{CAR}}(\mathfrak{h}) \cong M_{2^{d_h}}(\mathbb{C})$. $\square$

Corollary 3.2 (Existence of the Zini–Wang connecting unitaries). *In the situation of (1)–(2), an intertwining unitary $\phi_M^N : \mathcal{H}_M^{(N)} \rightarrow \mathcal{H}_{M+1}^{(N)}$ exists if and only if*

$$m(S_M^{(N)}) = m(S_{M+1}^{(N)}). \quad (7)$$

Purity of all the $\omega_M^{(N)}$, i.e. $m \equiv 0$, is sufficient for (7) but is not necessary.

Corollary 3.2 is the precise replacement for the manuscript’s “assuming a Stone–von Neumann-type result at finite scales”. It also shows that the inference in the second quotation of §1 — from non-purity of $\omega_M^{(N)}$ to the unavailability of the Zini–Wang structure — does not go through as stated: a family of mixed states of *constant* multiplicity supports connecting unitaries perfectly well. Section 5 identifies what does go wrong.

4 The model of the manuscript

We now compute m in each case. Recall from (60) the symbol of the massless lattice vacuum,

$$P(k) = \frac{1}{2}(\mathbb{1}_2 + \text{sign}(k)(-\sin(\frac{1}{2}\varepsilon_N k)\sigma_x + \cos(\frac{1}{2}\varepsilon_N k)\sigma_y)), \quad k \in \Gamma_{N,\pm}, \quad (8)$$

with $\text{sign}(0) = 0$, and from (129) that the massless scaling-limit symbol is $P_0(k) = \frac{1}{2}(\mathbb{1}_2 + \text{sign}(k)\sigma_y) = p_{\text{sign}(k)}$, i.e. the Hardy projection. We write $\varepsilon_{N,M} := \varepsilon_{N+M}$ for the lattice spacing at the fine scale.

Theorem 4.1 (Closed forms). *Let $\Gamma_{N,-}$ be the Neveu–Schwarz momentum lattice, so that $0 \notin \Gamma_{N,-}$. Then:*

(i) *$P(k)$ of (8) is a projection for every k , so the lattice vacuum is pure on the full algebra $\mathfrak{A}_{N,\pm}$;*

(ii) *with $p_\mp = \frac{1}{2}(\mathbb{1}_2 \mp \sigma_y)$,*

$$\|[P(k), p_\mp]\| = \frac{1}{2}|\sin(\frac{1}{2}\varepsilon_N k)|, \quad (9)$$

which vanishes if and only if $\varepsilon_N k \in 2\pi\mathbb{Z}$;

(iii) *the chiral compression $p_- P(k) p_-$ is the scalar*

$$\lambda_N(k) = \frac{1}{2}(1 - \text{sign}(k)\cos(\frac{1}{2}\varepsilon_N k)), \quad \lambda_N(1 - \lambda_N) = \frac{1}{4}\sin^2(\frac{1}{2}\varepsilon_N k); \quad (10)$$

(iv) *for the momentum-cutoff renormalization group of Definition 3.16 on the full algebra, $S_M^{(N)}(k) = P(k)|_{\varepsilon_N \rightarrow \varepsilon_{N,M}}$ for $k \in \Gamma_{N,\pm}$, which is a projection for every $M \in \mathbb{N}_0 \cup \{\infty\}$; hence $m(S_M^{(N)}) = 0$ throughout and $\omega_M^{(N)}$ is pure;*

(v) *for the momentum-cutoff renormalization group on the chiral algebra $\mathfrak{A}_{N,-}^{(-)}$,*

$$m(S_M^{(N)}) = |\Gamma_{N,-}| \quad \text{for every finite } M, \quad m(S_\infty^{(N)}) = 0; \quad (11)$$

(vi) for the wavelet renormalization group of Definition 3.2 on the chiral algebra, applied to the scaling-limit state, the symbol $(R_\infty^N)^* P^- R_\infty^N$ is circulant on $\ell^2(\Lambda_N)$ with eigenvalues

$$\mu_N(\kappa) = \sum_{j \in \mathbb{Z} : \kappa + 2\pi j / \varepsilon_N < 0} |\hat{s}(\varepsilon_N \kappa + 2\pi j)|^2, \quad \kappa \in \Gamma_{N,-}, \quad (12)$$

and these satisfy $\mu_N(\kappa) \in (0, 1)$ strictly, so $m = |\Gamma_{N,-}|$, together with

$$\min\{\mu_N(\kappa), 1 - \mu_N(\kappa)\} \leq \delta(\varepsilon_N \kappa) := 1 - |\hat{s}(\varepsilon_N \kappa)|^2 = \mathcal{O}((\varepsilon_N \kappa)^{2K}); \quad (13)$$

(vii) on the chiral algebra, for either renormalization group and every finite M , one has $m(S_M^{(N)}) = |\Gamma_{N,-}|$.

Proof. (i) The vector $\vec{n}(k) = \text{sign}(k)(-\sin(\frac{1}{2}\varepsilon_N k), \cos(\frac{1}{2}\varepsilon_N k), 0)$ is a unit vector for $k \neq 0$, and $\frac{1}{2}(\mathbb{1}_2 + \vec{n} \cdot \vec{\sigma})$ is a rank-one projection; for $k = 0$ (Ramond only) $P(0) = \frac{1}{2}\mathbb{1}_2$, cf. the text following (60).

(ii) Only the σ_x -part fails to commute with σ_y :

$$[P(k), p_\mp] = \mp \frac{1}{2} \cdot \frac{1}{2} \text{sign}(k) (-\sin(\frac{1}{2}\varepsilon_N k)) [\sigma_x, \sigma_y] = \pm \frac{i}{2} \text{sign}(k) \sin(\frac{1}{2}\varepsilon_N k) \sigma_z,$$

using $[\sigma_x, \sigma_y] = 2i\sigma_z$ and $[\sigma_y, \sigma_y] = 0$. Since $\|\sigma_z\| = 1$ and $|\text{sign}(k)| = 1$ for $k \neq 0$, (9) follows.

(iii) With $e_\pm = 2^{-1/2}(e_1 \pm ie_2)$ the eigenvectors of σ_y , one has $\sigma_y e_- = -e_-$ and $\sigma_x e_- = -ie_+ \perp e_-$, so $\langle e_-, P(k)e_- \rangle = \frac{1}{2}(1 - \text{sign}(k) \cos(\frac{1}{2}\varepsilon_N k))$. The second identity in (10) is elementary. Consistency with Lemma 2.1 is the computation $\|X\|^2 = \|[P, p_-]\|^2 = \frac{1}{4} \sin^2(\frac{1}{2}\varepsilon_N k) = \lambda_N(1 - \lambda_N) = \|S' - S'^2\|$.

(iv) By (121) the momentum-cutoff isometry acts as multiplication by $\varepsilon^{1/2}\chi_{\Gamma_N}$ on the Fourier side, so $\text{ran } R_{N+M}^N$ is the span of the momentum modes in $\Gamma_N \subset \Gamma_{N+M}$, with both $\mathbb{C}^2$ -components. The symbol $S^{(N+M)}$ acts fibrewise in k , hence leaves this span invariant; Lemma 2.1(iii) applies and the compression is the restriction of P to those fibres, still a projection. Letting $M \rightarrow \infty$ replaces P by P_0 , again a projection.

(v) By (iii) the compressed chiral symbol at $k \in \Gamma_{N,-}$ is the scalar $\lambda_{N+M}(k)$ of (10) with ε_N replaced by $\varepsilon_{N,M}$. It lies in $(0, 1)$ unless $\sin(\frac{1}{2}\varepsilon_{N,M} k) = 0$, i.e. unless $\varepsilon_{N,M} k \in 2\pi\mathbb{Z}$. For $k \in \Gamma_{N,-}$ we have $|\varepsilon_{N,M} k| \leq \varepsilon_{N,M}\pi/\varepsilon_N = 2^{-M}\pi < 2\pi$, and $\varepsilon_{N,M} k = 0$ is excluded because $0 \notin \Gamma_{N,-}$. Hence every one of the $|\Gamma_{N,-}|$ eigenvalues lies strictly in $(0, 1)$, for every finite M . As $M \rightarrow \infty$, $\lambda_{N+M}(k) \rightarrow \frac{1}{2}(1 - \text{sign}(k)) = \chi_{k < 0} \in \{0, 1\}$, so m drops to 0.

(vi) By Definition 3.2 the wavelet isometry acts on the Fourier side as multiplication by $\varepsilon_N^{1/2}\hat{s}(\varepsilon_N \cdot)$ followed by periodisation, so for $\xi, \eta \in \ell^2(\Lambda_N)$

$$\langle \xi, (R_\infty^N)^* P^- R_\infty^N \eta \rangle = \frac{1}{|\Lambda_N|} \sum_{l \in \Gamma_{\infty,-}, l < 0} |\hat{s}(\varepsilon_N l)|^2 \hat{\xi}_l \hat{\eta}_l,$$

which is invariant under simultaneous translation of ξ, η on Λ_N , hence circulant; its eigenvalue at the lattice momentum $\kappa \in \Gamma_{N,-}$ is the sum of $|\hat{s}|^2$ over the negative members of the fibre $\kappa + \frac{2\pi}{\varepsilon_N}\mathbb{Z}$, which is (12). Orthonormality of $\{s(\cdot - n)\}_{n \in \mathbb{Z}}$ is equivalent to $\sum_{j \in \mathbb{Z}} |\hat{s}(\xi + 2\pi j)|^2 = 1$, so the full fibre sums to 1 and

$$1 - \mu_N(\kappa) = \sum_{j : \kappa + 2\pi j / \varepsilon_N > 0} |\hat{s}(\varepsilon_N \kappa + 2\pi j)|^2.$$

If $\kappa < 0$ the term $j = 0$ occurs in μ_N , whence $1 - \mu_N(\kappa) \leq \sum_{j \neq 0} |\hat{s}(\varepsilon_N \kappa + 2\pi j)|^2 = 1 - |\hat{s}(\varepsilon_N \kappa)|^2 = \delta(\varepsilon_N \kappa)$; if $\kappa > 0$ the same argument bounds $\mu_N(\kappa)$. This is (13), and $\delta(\xi) = \mathcal{O}(\xi^{2K})$ is (249) of the manuscript. Strict positivity of both μ_N and $1 - \mu_N$ holds because $\hat{s}$ vanishes only on $2\pi\mathbb{Z} \setminus \{0\}$ while $\varepsilon_N \kappa + 2\pi j \in 2\pi\mathbb{Z}$ would force $\varepsilon_N \kappa \in 2\pi\mathbb{Z}$, excluded as in (v). Indeed the single

term $j = -1$ (for $\kappa > 0$) gives the matching lower bound $\mu_N(\kappa) \geq |\hat{s}(\varepsilon_N \kappa - 2\pi)|^2 \geq c(\varepsilon_N \kappa)^{2K}$, since $\hat{s}$ has a zero of order exactly K at -2π .

(vii) Both renormalization groups act on the Fourier side by multiplication with a *scalar* symbol — $\varepsilon^{1/2} \chi_{\Gamma_N}$ in the momentum-cutoff case (121), $\varepsilon_N^{1/2} \hat{s}(\varepsilon_N \cdot)$ in the wavelet case — hence commute with the constant spinor projections $p_{\pm}$ and restrict to the chiral one-particle space. By (iii) and (v) the chiral symbol at the fine scale is the multiplication operator λ_{N+M} , which satisfies $0 < \lambda_{N+M} < 1$ strictly on $\Gamma_{N+M,-}$; (4) of Lemma 2.1 therefore gives $m(S_M^{(N)}) = \dim \mathfrak{h}_{N,-}^{(-)} = |\Gamma_{N,-}|$, whichever isometry is used. $\square$

Remark 4.2 (The aliasing mass is the same object twice). The quantity $\delta(\xi) = 1 - |\hat{s}(\xi)|^2 = \sum_{j \neq 0} |\hat{s}(\xi - 2\pi j)|^2$ controlling (13) is exactly the aliasing mass that drives the Jackson estimate of Lemma 4.28 in the manuscript, and its order $2K$ is (249). The failure of the wavelet renormalization group to produce pure finitely renormalized states is therefore the *same* phenomenon as the failure of $P^{\pm}$ to preserve the wavelet core $\mathcal{D}_W$ recorded in Remark 4.22: both say that a multiresolution subspace is not a spectral subspace of a Fourier multiplier, and both are quantified by δ .

5 The obstruction, and its size

Corollary 5.1 (Where the Zini–Wang structure really fails). *For the chiral algebra $\mathfrak{A}_{N,-}^{(-)}$ with the momentum-cutoff renormalization group,*

$$\dim \mathcal{H}_M^{(N)} = 2^{|\Gamma_{N,-}|} \cdot 2^{|\Gamma_{N,-}|} \text{ for every finite } M, \quad \dim \mathcal{H}_{\infty}^{(N)} = 2^{|\Gamma_{N,-}|}. \quad (14)$$

Consequently:

- (i) the connecting unitaries $\phi_M^N : \mathcal{H}_M^{(N)} \rightarrow \mathcal{H}_{M+1}^{(N)}$ of the manuscript do exist, for every finite M , by Corollary 3.2 — notwithstanding that every $\omega_M^{(N)}$ is mixed;
- (ii) but no such unitary exists between any $\mathcal{H}_M^{(N)}$ and $\mathcal{H}_{\infty}^{(N)}$: the GNS multiplicity collapses by the factor $2^{|\Gamma_{N,-}|}$ in the limit.

Hence the inductive limit of the $\mathcal{H}_M^{(N)}$ along the ϕ_M^N is not the scaling-limit representation. Two comparisons are worth drawing. On the full algebra $\mathfrak{A}_{N,\pm}$, Theorem 4.1(iv) gives $m \equiv 0$ throughout and no collapse occurs. On the chiral algebra with the wavelet renormalization group, Theorem 4.1(vi),(vii) gives $m = |\Gamma_{N,-}|$ at every M including $M = \infty$, so again no collapse occurs: the wavelet route keeps a mixed limit state and with it a constant multiplicity. The collapse is thus specific to the momentum-cutoff route, whose limit state is pure precisely because its isometries are spectral projections of the same multiplier that defines the Hardy projection.

Proof. Immediate from Proposition 3.1 and Theorem 4.1(iv),(v), with $d_{\mathfrak{h}} = |\Gamma_{N,-}|$ for the chiral one-particle space. $\square$

This is the precise form of the manuscript’s assertion, and it differs from it in an instructive way. The obstruction is not that the finitely renormalized states fail to be pure — they do fail, but that alone is harmless. The obstruction is that they fail to be pure *by an amount that does not survive the limit*: the multiplicity is locally constant in M and then jumps at $M = \infty$. The manuscript’s parenthetical “(due to the entanglement of the chiral halves)” is vindicated as the correct *mechanism* — by Theorem 4.1(ii) the chiral halves are entangled exactly to the extent that $[P(k), p_{\mp}] \neq 0$ — while “not pure” is the wrong *criterion*.

Corollary 5.2 (The mixing is extensive, with density $2 \ln 2 - 1$). *The chiral restriction of the finite-scale massless lattice vacuum has von Neumann entropy*

$$S(\omega_0^{(N)}|_{\mathfrak{A}_{N,-}^{(-)}}) = \sum_{k \in \Gamma_{N,-}} H_2(\lambda_N(k)), \quad H_2(p) := -p \ln p - (1-p) \ln(1-p), \quad (15)$$

with λ_N as in (10), and the entropy density converges:

$$\lim_{N \rightarrow \infty} \frac{1}{|\Gamma_{N,-}|} S(\omega_0^{(N)}|_{\mathfrak{A}_{N,-}^{(-)}}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_2\left(\frac{1 - \cos(u/2)}{2}\right) du = 2 \ln 2 - 1 \approx 0.386294. \quad (16)$$

The mixing is concentrated at the Brillouin-zone boundary, where $\lambda_N(1 - \lambda_N) \rightarrow \frac{1}{4}$ is maximal, and vanishes as $\frac{1}{4}(\frac{1}{2}\varepsilon_N k)^2 + \mathcal{O}((\varepsilon_N k)^4)$ on the modes that survive the scaling limit.

Proof. For a quasi-free state the entropy is $\sum_i H_2(\lambda_i)$ over the eigenvalues of the symbol, which with (10) is (15). Writing $u = \varepsilon_N k$, the momenta $k \in \Gamma_{N,-}$ equidistribute in $u \in (-\pi, \pi)$ as $N \rightarrow \infty$, and $\lambda_N = \frac{1}{2}(1 - \text{sign}(u) \cos(u/2))$, whose binary entropy is even in u and equals $H_2(\frac{1}{2}(1 - \cos(u/2)))$; this gives the Riemann sum in (16). For the value, substitute $u = 2t$ and then $t = 2\varphi$, and use $H_2(\sin^2 \varphi) = -2 \sin^2 \varphi \ln \sin \varphi - 2 \cos^2 \varphi \ln \cos \varphi$:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} H_2\left(\frac{1 - \cos(u/2)}{2}\right) du = \frac{2}{\pi} \int_0^{\pi/2} H_2(\sin^2(t/2)) dt = -\frac{8}{\pi} \int_0^{\pi/4} [\sin^2 \varphi \ln \sin \varphi + \cos^2 \varphi \ln \cos \varphi] d\varphi.$$

The substitution $\varphi \mapsto \frac{\pi}{2} - \varphi$ turns the second summand into the first over $[\frac{\pi}{4}, \frac{\pi}{2}]$, so the bracket integrates to $\int_0^{\pi/2} \sin^2 \varphi \ln \sin \varphi d\varphi = \frac{\pi}{8}(1 - \ln 4)$, whence the value $-\frac{8}{\pi} \cdot \frac{\pi}{8}(1 - 2 \ln 2) = 2 \ln 2 - 1$. $\square$

So the chiral restriction is not merely mixed: it is mixed at maximal rate on a finite fraction of the modes, and its entropy grows like $(2 \ln 2 - 1)|\Gamma_{N,-}|$. What makes the scaling limit pure nevertheless is that the strongly mixed modes are precisely the lattice artefacts near $|k| \sim \pi/\varepsilon_N$, which are pushed to infinity by the limit, while each fixed mode is mixed only to order $(\varepsilon_N k)^2$.

6 What survives unconditionally

For completeness we record the part of the comparison that requires no hypotheses, and which the manuscript states correctly.

Proposition 6.1 (OAR supplies Zini–Wang-compatible data). *Let $\{\mathfrak{A}_N, \alpha_{N+1}^N\}$ be an OAR inductive system with a projectively consistent family of scaling-limit states $\omega_\infty^{(N)}$, and let $\{\mathcal{H}_\infty^{(N)}, \pi_\infty^{(N)}, \Omega_\infty^{(N)}\}$ be the GNS data. Then there are isometries $V_{N+1}^N : \mathcal{H}_\infty^{(N)} \rightarrow \mathcal{H}_\infty^{(N+1)}$ with $V_{N+1}^N \Omega_\infty^{(N)} = \Omega_\infty^{(N+1)}$ and*

$$V_{N+1}^N \pi_\infty^{(N)}(O) = \pi_\infty^{(N+1)}(\alpha_{N+1}^N(O)) V_{N+1}^N, \quad O \in \mathfrak{A}_N, \quad (17)$$

and consequently

$$V_{N+1}^N \pi_\infty^{(N)}(O) (V_{N+1}^N)^* = \pi_\infty^{(N+1)}(\alpha_{N+1}^N(O)) p_{\text{ran } V_{N+1}^N}. \quad (18)$$

Moreover V_{N+1}^N is unitary if and only if $\pi_\infty^{(N+1)}(\alpha_{N+1}^N(\mathfrak{A}_N)) \Omega_\infty^{(N+1)}$ is dense in $\mathcal{H}_\infty^{(N+1)}$. In the quasi-free case with $\omega_\infty^{(N)}$ pure, V_{N+1}^N is the second quantisation $\Gamma(R_{N+1}^N)$ of the one-particle isometry, which is exactly the shape of a Zini–Wang strong scaling limit.

Proof. Projective consistency $\omega_\infty^{(N+1)} \circ \alpha_{N+1}^N = \omega_\infty^{(N)}$ means that $\pi_\infty^{(N)}(O)\Omega_\infty^{(N)} \mapsto \pi_\infty^{(N+1)}(\alpha_{N+1}^N(O))\Omega_\infty^{(N+1)}$ preserves inner products, hence extends to an isometry V_{N+1}^N on the closure, which is (17) by construction; (18) follows by multiplying (17) with $(V_{N+1}^N)^*$ on the right and using $V^*V = \mathbb{1}$, $VV^* = p_{\text{ran } V}$. The range of V_{N+1}^N is by construction the closure of $\pi_\infty^{(N+1)}(\alpha_{N+1}^N(\mathfrak{A}_N))\Omega_\infty^{(N+1)}$, giving the unitarity criterion. The last statement is the standard identification of the GNS representation of a pure quasi-free state with the Fock representation over the one-particle space. $\square$

Equation (18) is the identity displayed in §1.2 of the manuscript; Proposition 6.1 supplies its proof and the accompanying unitarity criterion.

7 Summary

	$m(S_M^{(N)}), M < \infty$	$m(S_\infty^{(N)})$	ϕ_M^N exist?	reaches the limit?
full algebra, mom. cutoff	0	0	yes	yes
chiral algebra, mom. cutoff	$ \Gamma_{N,-} $	0	yes	no
chiral algebra, wavelet	$ \Gamma_{N,-} $	$ \Gamma_{N,-} $	yes	yes

Only the first row consists of pure states; rows two and three are mixed throughout, and mixedness is visibly not what decides the last column.

The three conclusions, in order of importance:

- (1) The criterion for the Zini–Wang connecting unitaries is constancy of the GNS multiplicity m , not purity (Corollary 3.2). The manuscript’s hypothesis “assuming a Stone–von Neumann-type result at finite scales” is exactly (7).
- (2) For the chiral algebra the multiplicity is constant at every finite scale and collapses only in the limit, so the obstruction is a discontinuity at $M = \infty$, not the mixedness of the $\omega_M^{(N)}$ (Corollary 5.1); and the collapse is specific to the momentum-cutoff route, the wavelet route having constant multiplicity all the way to $M = \infty$. The mechanism named in the manuscript — the entanglement of the chiral halves — is correct, and is quantified by $\|[P(k), p_\mp]\| = \frac{1}{2}|\sin(\frac{1}{2}\varepsilon_N k)|$.
- (3) The failure of purity is extensive, with entropy density exactly $2 \ln 2 - 1$ per site (Corollary 5.2); it is carried by the zone-boundary lattice artefacts and is of order $(\varepsilon_N k)^2$ on the modes that survive.

8 What this note does not settle

- *The converse direction.* We have shown that OAR supplies Zini–Wang data (Proposition 6.1) and determined exactly when the connecting unitaries exist. We have not shown that a Zini–Wang low-energy scaling limit conversely determines an OAR inductive system; this would require reconstructing the refining morphisms α from the ϕ ’s, and the identity (18) shows that the ϕ ’s see only the compression of $\pi(\mathfrak{A}_N)$ to $\text{ran } V$.
- *Beyond quasi-free.* Proposition 3.1 holds for arbitrary states on a finite-dimensional algebra, but the closed forms of Theorem 4.1 use quasi-freeness throughout. The Zini–Wang conjecture that higher-level anyonic chains admit compatible embeddings [5, p. 898] is untouched by these methods.

- *Infinite scales.* All statements are at fixed finite N , where $\mathfrak{A}_N$ is a matrix algebra. The passage $N \rightarrow \infty$ leaves the setting of Proposition 3.1, and the relevant criterion there is quasi-equivalence of quasi-free states, governed by the Powers–Størmer condition that $S^{1/2} - S'^{1/2}$ be Hilbert–Schmidt [1, 4]. We have not carried out that analysis.
- *The strong scaling limit.* Zini and Wang’s strong scaling limit requires the ϕ ’s to come from a single isometry $\phi_N : \mathcal{H}_N \rightarrow \mathcal{H}_{N+1}$ of the full spaces. Proposition 6.1 produces such an isometry in the pure quasi-free case, but we have not examined whether the mixed case of Theorem 4.1(v)–(vii) admits one.

Numerical verification

All closed forms above are checked in `numerics/os_check19.py` (pure `numpy`):

- [31] $\|P^2 - P\| \leq 1.1 \cdot 10^{-16}$; (9) to machine zero (0.0); (10) to $3.3 \cdot 10^{-16}$; the entropy density of (16) agrees with $2 \ln 2 - 1$ to 10 decimal places.
- [32] the momentum-cutoff compression satisfies $\|S'^2 - S'\| \leq 7.9 \cdot 10^{-17}$ for all tested N, M — Theorem 4.1(iv).
- [33] the circulant spectrum (12) reproduces the directly computed eigenvalues to 10^{-15} (db4, db6) and the bound (13) holds at every κ . For db2 the agreement is only 10^{-10} , a replica-truncation artefact. Note that $\min\{\mu, 1 - \mu\}$ can be as small as 10^{-12} , so the strict positivity asserted in Theorem 4.1(vi) is at the edge of double precision and is established analytically, not numerically.
- [34] the multiplicities of (11): $m = 0$ throughout on the full algebra; $m = |\Gamma_{N,-}|$ for $M = 1, \dots, 7$ on the chiral algebra and $m = 0$ at $M = \infty$, for $N = 3, 5$.

A numerically stable evaluation of $\hat{s}$ is required for [33]: the order- K zero of m_0 at π is deflated in closed form, since evaluating $m_0(\pi + \eta) \sim \eta^K$ from the filter coefficients loses all precision.

References

- [1] H. Araki. On quasifree states of CAR and Bogoliubov automorphisms. *Publ. Res. Inst. Math. Sci.* **6** (1970/71), 385–442.
- [2] D. E. Evans and Y. Kawahigashi. *Quantum Symmetries on Operator Algebras*. Oxford University Press, 1998.
- [3] T. J. Osborne and A. Stottmeister. Conformal field theory from lattice fermions. *Comm. Math. Phys.* **398** (2023), 219–289; arXiv:2107.13834. Referenced here in the revised form `free_fermion_cft_v5.tex`.
- [4] R. T. Powers and E. Størmer. Free states of the canonical anticommutation relations. *Comm. Math. Phys.* **16** (1970), 1–33.
- [5] M. S. Zini and Z. Wang. Conformal field theories as scaling limit of anyonic chains. *Comm. Math. Phys.* **363** (2018), 877–953.